

Theorem (Chebyshev's Sum Inequality). *Let*

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. Then

$$\boxed{n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right)}.$$

Proof. 1. **Degenerate cases.**

- $n = 0$: the statement is vacuously true.
- $n = 1$: the inequality reads $a_1 b_1 \geq a_1 b_1$, true with equality.
- If the a -sequence (or the b -sequence) is constant, each factor $(a_i - a_j)$ (resp. $(b_i - b_j)$) is 0, so the two sides are equal. These are the trivial equality cases.

Hence we may assume $n \geq 2$ and that at least one of the sequences is not constant.

2. **A non-negative double sum.** Define

$$S := \sum_{1 \leq i < j \leq n} (a_i - a_j)(b_i - b_j).$$

Because the sequences are non-decreasing, for $i < j$ we have $a_i - a_j \leq 0$ and $b_i - b_j \leq 0$, whence

$$(a_i - a_j)(b_i - b_j) \geq 0.$$

Summing over all $\binom{n}{2}$ pairs gives

$$S \geq 0. \tag{1}$$

3. **Expansion of S .** Write the sum symmetrically:

$$S = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i - a_j)(b_i - b_j).$$

Expanding the product inside the double sum,

$$\begin{aligned} S &= \frac{1}{2} \left(\sum_i \sum_j a_i b_i - \sum_i \sum_j a_i b_j - \sum_i \sum_j a_j b_i + \sum_i \sum_j a_j b_j \right) \\ &= \frac{1}{2} \left(n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right) - \left(\sum_{j=1}^n a_j \right) \left(\sum_{i=1}^n b_i \right) + n \sum_{i=1}^n a_i b_i \right) \\ &= n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \end{aligned}$$

Thus we have the identity

$$\sum_{1 \leq i < j \leq n} (a_i - a_j)(b_i - b_j) = n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \quad (2)$$

4. **Deriving the inequality.** From (1) and (2) we obtain

$$n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \geq 0,$$

which is exactly the desired inequality.

5. **Equality condition.** Equality holds iff $S = 0$. Since each term $(a_i - a_j)(b_i - b_j)$ is non-negative, $S = 0$ precisely when every term vanishes, i.e.

$$(a_i - a_j)(b_i - b_j) = 0 \quad \text{for all } 1 \leq i < j \leq n.$$

Hence for each pair (i, j) we must have either $a_i = a_j$ or $b_i = b_j$. Consequently the equality cases are:

- (a) All a_i are equal (the a -sequence is constant);
- (b) All b_i are equal (the b -sequence is constant);
- (c) More generally, the index set $\{1, \dots, n\}$ can be partitioned into blocks such that on each block the a 's are equal and, on the same block, the b 's are equal.

When both sequences are strictly increasing, only the first two possibilities can occur.

□